

Python Programming (B.Tech CSE - Sem 2)

Experiment 01 – Exercise Sheet

Python Installation and Basic Python Statements

Tofik Ali

February 20, 2026

Repository: <https://github.com/tali7c/Python-Programming>

Note: This document contains only problem statements (no solutions).

Instructions

- Write a separate Python program for each programming exercise.
- Use meaningful variable names and clear output formatting.
- Validate input where appropriate (example: non-negative values).

Exercises

Exercise 00: Python Installation and Modes (Non-coding)

- Install Python 3.
- Write the steps of installation.
- Explain the difference between interactive mode and scripting mode (IDLE/terminal).

Exercise 01: Age Type

Create a variable to store your age and print its type using `type()`.

Exercise 02: Hello String

Declare a string variable called `x` and assign it the value `"Hello"`. Print the value of `x`.

Exercise 03: Print Different Data Types

Take different data types (int, float, string, boolean, etc.) and print values using the `print` function. Also print their types.

Exercise 04: Integer Arithmetic

Declare `x=9` and `y=7`. Perform addition, multiplication, division and subtraction on these two variables and print the results.

Exercise 05: Hypotenuse (Pythagoras Theorem)

Write a program to compute the length of the hypotenuse (c) of a right triangle using Pythagoras theorem.

Exercise 06: Simple Interest

Write a program to find simple interest.

Exercise 07: Triangle Area (Sides Given)

Write a program to find area of a triangle when lengths of sides are given. Validate whether the sides form a valid triangle.

Exercise 08: Seconds Conversion

Convert given seconds into hours, minutes and remaining seconds.

Exercise 09: Swap Without Extra Variable

Swap two numbers without taking an additional variable.

Exercise 10: Sum of First n Natural Numbers

Find sum of first n natural numbers.

Exercise 11: Truth Table for Bitwise Operators

Print truth table for bitwise operators `&`, `|` and `^` for inputs $a, b \in \{0, 1\}$.

Exercise 12: Left Shift and Right Shift

Find left shift and right shift values of a given number.

Exercise 13: Membership Operator

Using membership operator, find whether a given number is in sequence (10, 20, 56, 78, 89).